

A Unified Mathematical Framework for Substrate-Agnostic Emergent Systems

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Abstract—Research on emergent systems spans artificial life, open-ended evolution, multi-agent reinforcement learning, cultural evolution, active inference, and causal-emergence theory. These literatures often study related phenomena using incompatible mathematical objects, which makes comparison across substrates and evaluation regimes difficult. This paper proposes a substrate-agnostic mathematical framework organized around five typed components: substrate, variation, viability, interaction topology, and observer-evaluator. The framework specifies how these components compose, how entities can be defined through support sets and scale projections, and how Hoel-style causal emergence can be posed as an assessment problem for the resulting systems. The framework is illustrated with representative instances from cellular automata, continuous artificial-life fields, program-based evolution, multi-agent systems, and foundation-model-mediated search, followed by a worked composite example and implementation checklist. The paper also separates established definitions from conjectural unifications and states open problems whose resolution would determine the framework’s scope. The result is a methodology for specifying, comparing, and implementing emergent-system models without assuming a single substrate, fitness landscape, or observer.

Index Terms—Artificial life, causal emergence, open-ended evolution, substrate-agnostic systems, viability.

I. INTRODUCTION

Emergent-system research has produced a broad range of models, including cellular automata, continuous artificial-life substrates, digital organisms, multi-agent learning systems, cultural-evolution models, and active-inference architectures. Although these systems differ substantially in implementation and interpretation, many of them can be described in terms of five recurring roles: a substrate, a variation process, a viability criterion, an interaction topology, and an observer or evaluator. This paper gives those roles a common mathematical typing and studies the resulting composition rules.

The proposed framework is intentionally non-prescriptive. It specifies the kinds of objects that occupy each role and the ways in which the roles compose, but it does not impose a particular fitness landscape, topology, observer, or theory of agency. Its purpose is therefore comparative and constructive: comparative because it provides a common vocabulary for existing traditions, and constructive because it identifies reusable modules for building composite systems.

The main contributions are as follows. First, the paper defines a substrate-agnostic emergent system as a typed tuple with an explicit iteration rule. Second, it introduces composition operators for heterogeneous substrates and coupled variation channels. Third, it proposes an information-theoretic

entity definition that separates physical support from scale of description. Fourth, it positions viability, observer-relative novelty, and causal emergence as related but distinct assessment problems. Finally, it records the principal conjectures and open problems rather than presenting them as settled results.

II. BACKGROUND AND RELATED WORK

Emergent-system research spans several traditions that often share modelling roles while differing in notation and substrate. Cellular automata and continuous artificial-life systems emphasize substrate dynamics and local interaction rules; digital-organism and open-ended-evolution systems emphasize variation, selection, and persistent novelty; multi-agent and cultural-evolution models emphasize topology-mediated transmission; and active-inference models emphasize viability-like constraints expressed through Markov blankets and free-energy parameterisations.

Emergence itself remains contested. In artificial life, Gershenson argues for an information-based view in which emergent structure is information present at one scale but absent at another [1]. Hoel’s causal emergence programme gives a more specific criterion: a macro-description is emergent when it has greater effective information than an appropriate micro-description [2], [3]. These views are compatible with the present paper’s separation between substrate dynamics and assessment: the substrate generates trajectories, while an observer or causal analysis decides whether a higher-scale description carries new information.

Open-ended evolution adds a second pressure on any framework for emergence: the system should not merely produce a single surprising pattern, but should continue to generate meaningful novelty. Classic novelty search made this operational by rewarding behavioral distance from an archive rather than direct objective improvement [4]. Recent analysis clarifies that novelty search is best understood at the level of archive coverage and saturation, rather than as a purely objectiveless process [5]. Quality-Diversity algorithms extend this idea by maintaining repertoires of diverse high-performing solutions; recent Lenia work uses this approach to discover diverse self-organizing patterns in a continuous cellular-automaton substrate [6]. Novelty can also be used as an auxiliary objective for unbounded search spaces, where it helps identify promising regions before local optimization [7].

Open-endedness is also broader than genetic evolution. Cultural evolution has been proposed as a complementary case in which cumulative modification, affordance discovery, and exaptation generate new kinds of novelty [8]. Abiotic evolving systems provide useful counterpoints: Hazen and

Wong analyze mineral evolution as a case where functional information increases but appears bounded, sharpening the distinction between bounded diversity growth and genuinely open-ended evolution [9]. Stepney’s recent overview of virtual artificial life likewise emphasizes that artificial life spans many substrates, including synthetic, robotic, wetware, and purely computational systems, and that no single implementation mechanism exhausts the design space [10].

Large language models and foundation models introduce a newer route to variation and evaluation. LLM-based genetic programming uses a language model inside evolutionary operators for code generation and mutation [11]; related foundation-model-guided exploration methods use learned priors or embedding spaces to bias search toward semantically interesting regions [12]–[14]. These systems make the observer–evaluator slot especially important: novelty may be measured not only in physical state space or behavior space, but also in learned representational spaces.

This paper is closest in spirit to four families of prior work. First, the algebra of open dynamical systems [15], [16] provides compositional tools for wiring dynamical systems together. Second, information-theoretic individuality [17] motivates the separation between physical support and scale of description. Third, novelty-and-learnability accounts of open-endedness [18] motivate observer-relative evaluation. Fourth, Hoel-style causal emergence provides a quantitative emergence criterion based on effective information across scales. The present contribution is not a replacement for those traditions. It is a typing and composition scaffold that makes their recurring roles explicit: substrate, variation, viability, interaction topology, and observer–evaluator. The framework therefore treats dynamical-regime descriptors such as Wolfram classes, edge-of-chaos behavior, and self-organized criticality as substrate-side properties, while treating causal emergence and observer-relative novelty as assessment problems over trajectories.

III. METHODOLOGY

A. Notation and conventions

Throughout, $\mathbb{N}, \mathbb{R}_{\geq 0}, \mathbb{R}$ have their usual meanings. Table I collects the notation used across multiple sections; symbols introduced only locally are defined where they appear. Composition is written $\circ$, product measure spaces use $\otimes$, and disjoint unions use $\sqcup$. Discrete time indices are $t \in \mathbb{N}$; population indices are $i \in \{1, \dots, N\}$.

Given a partially ordered set $(P, \sqsubseteq)$, a *closure operator* is a map $c : 2^P \rightarrow 2^P$ satisfying (i) extensivity $A \subseteq c(A)$, (ii) monotonicity $A \subseteq B \Rightarrow c(A) \subseteq c(B)$, and (iii) idempotence $c(c(A)) = c(A)$. A set A is *closed* iff $A = c(A)$.

B. Proposed System Definition

A **substrate-agnostic emergent system** is a tuple

$$\mathcal{S} = (\mathbf{X}, V, F, T, O),$$

where the substrate $\mathbf{X} = (X, \Phi, \mu_0)$ provides the state space X (§III-C), and V, F, T, O are the variation, viability, interaction-

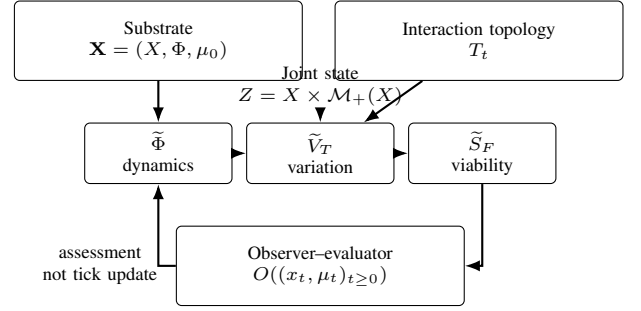


Fig. 1. The five-slot model. The substrate supplies state and dynamics; interaction topology mediates variation; viability filters the resulting population measure; and the observer–evaluator assesses trajectories rather than altering the default tick update.

topology, and observer operators respectively. The system evolves on the *joint state space*

$$Z := X \times \mathcal{M}_+(X),$$

whose elements (x, μ) pair a substrate state $x \in X$ with a population measure $\mu \in \mathcal{M}_+(X)$ over the entities (§III-D) currently inhabiting x . Each slot induces a lifted operator on Z :

$$\begin{aligned} \tilde{\Phi} : Z &\rightarrow Z, & (x, \mu) &\mapsto (\Phi(x), \Phi_*\mu), \\ \tilde{V}_T : Z &\rightarrow Z, & (x, \mu) &\mapsto (x, V_T(\mu; x)), \\ \tilde{S}_F : Z &\rightarrow Z, & (x, \mu) &\mapsto (x, F(\mu)), \end{aligned}$$

where $\Phi_*\mu$ is the push-forward of μ along Φ (the identity in substrate-as-population paradigms in which the population is implicit in x); $V_T(\mu; x)$ is the population-level variation kernel induced by V under topology T_t , formalised in §III-E.1; and F is the viability filter of §III-F. The **iteration rule** at each tick $t \mapsto t + 1$ is

$$(x_{t+1}, \mu_{t+1}) = (\tilde{S}_F \circ \tilde{V}_T \circ \tilde{\Phi})(x_t, \mu_t).$$

The observer O acts on trajectories $(x_t, \mu_t)_{t \geq 0}$ rather than on single ticks. The slots are formalised in §III-C and §§III-E–III-H; entities in §III-D; and emergence in §III-I. The choice of iteration order is itself a modelling commitment; alternative orderings are discussed in §V-B.

In less compressed terms, $\mathcal{S}$ is the recipe for a simulation. The substrate is the material being simulated: a lattice of cells, a continuous field, a swarm of particles, a population of programs, or a coupled product of such materials. Variation proposes changes to the population living on that material; viability decides which proposed entities remain; topology says who can affect whom; and the observer records what the resulting trajectory looks like. The mathematics below fixes the types of those choices so that two systems can be compared without quietly changing the meaning of a slot.

Computational implication. The product structure is deliberate: each slot has its own typing and can be swapped without rewriting the others. In §III-L we argue that this same modularity supports a natural GPU mapping in which each slot becomes a parallel kernel.

TABLE I
RECURRING NOTATION USED THROUGHOUT THE FRAMEWORK.

Symbol	Name	Role in the paper
$\bar{S}$	Scaffolded system	Full substrate-agnostic emergent system.
$\mathbf{X} = (X, \Phi, \mu_0)$	Substrate	State space, substrate dynamics, and initial population measure.
X	State space	Underlying measurable space on which substrate states live.
x_t	Substrate state	State of X at tick t .
$\mu_t \in \mathcal{M}_+(X)$	Population measure	Finite positive measure representing entities inhabiting the substrate.
$Z = X \times \mathcal{M}_+(X)$	Joint state space	State space for the lifted tick update.
Φ	Substrate dynamics	Updates substrate state; stochastic variants map into $\mathcal{P}(X)$.
V / V_T	Variation operator	Generates variants; V_T is the population-level kernel mediated by topology.
F / S_F	Viability filter	Selects or reweights the post-variation population.
T_t	Interaction topology	Hypergraph or graph over current entities at tick t .
O / O_s	Observer–evaluator	Scores trajectories; O_s additionally carries observer state such as an archive.
$\tilde{\Phi}, \tilde{V}_T, \tilde{S}_F$	Lifted operators	Slot operators lifted to act on Z .
$\mathcal{E}_t$	Entity set	Entities detected at time t .
$e = (S, \pi)$	Entity	Supporting set S plus scale projection π .
π	Scale projection	Map from supported microstate to entity-level description.
$\boxtimes_\kappa$	Substrate composition	Coupled composition of heterogeneous substrates using coupling datum κ .
κ	Substrate coupling	Cross-substrate map that re-parameterises component dynamics.
$\diamond_\rho$	Variation-channel composition	Coupled composition of variation channels using coupling rule ρ .
ρ	Variation coupling	Rule coupling one variation channel’s sampling to another.
$\mathcal{P}(Y)$	Probability measures	Probability measures on measurable space Y .
$\mathcal{M}_+(Y)$	Finite positive measures	Finite non-negative measures on measurable space Y .
Π	Coarse-graining	Partition or quotient used in causal-emergence assessment.
$\text{EI}(M)$	Effective information	Hoel-style effective information of a Markov chain M .
$\text{CE}(M, \Pi)$	Causal emergence	Difference $\text{EI}(M^\Pi) - \text{EI}(M)$ at coarse-graining Π .
Ω	Open-endedness metric	Residence-time/open-endedness metric used by some observers.

C. Substrates

A **substrate** is a tuple

$$\mathbf{X} = (X, \Phi, \mu_0),$$

where X is a measurable space (the *substrate state space*), $\Phi : X \rightarrow X$ (or $\Phi : X \rightarrow \mathcal{P}(X)$ in the stochastic case) is a measurable *substrate dynamics* map, and $\mu_0 \in \mathcal{M}_+(X)$ is an initial measure. Plainly, the substrate is the simulation material plus its native update rule: the grid and cellular-automaton update, the Flow-Lenia field and its kernel dynamics, the particle state and its motion law, or the program space and interpreter dynamics. We deliberately *exclude* a “neighborhood structure” from the substrate tuple; the relationship between the substrate and the interaction topology is treated in §III-G.

Substrate composition operator $\boxtimes$.

Real instances are often heterogeneous: a Lenia field coupled to a meteorological forcing field coupled to a population of LLM agents. A naïve Cartesian product $X_1 \times X_2$ inherits no shared geometry, no shared time-scale, and no notion of cross-substrate coupling. We therefore introduce the **substrate composition operator** $\boxtimes$ in two flavours.

Definition III.1 (Free composition). Given substrates $\mathbf{X}_1, \mathbf{X}_2$, their *free composition* $\mathbf{X}_1 \boxtimes \mathbf{X}_2$ has state space $X_1 \times X_2$, dynamics $\Phi_1 \times \Phi_2$, and initial measure $\mu_0^{(1)} \otimes \mu_0^{(2)}$. This recovers the Cartesian product as a degenerate case (no coupling).

Definition III.2 (Coupled composition). A *coupling datum* between $\mathbf{X}_1$ and $\mathbf{X}_2$ is a pair of measurable maps $\kappa = (\kappa_{1 \rightarrow 2}, \kappa_{2 \rightarrow 1})$ with $\kappa_{i \rightarrow j} : X_i \rightarrow \text{Param}(\Phi_j)$, where $\text{Param}(\Phi_j)$ is a measurable space of parameters that re-shape the dynamics Φ_j . The κ -coupled composition $\mathbf{X}_1 \boxtimes_\kappa \mathbf{X}_2$ has state space $X_1 \times X_2$, dynamics

$$\tilde{\Phi}(x_1, x_2) = (\Phi_1[\kappa_{2 \rightarrow 1}(x_2)](x_1), \Phi_2[\kappa_{1 \rightarrow 2}(x_1)](x_2)),$$

and initial measure inherited as in Definition III.1.

Proposition III.3 (Associativity up to canonical isomorphism). *Free composition is symmetric monoidal on the category of substrates and substrate morphisms; coupled composition is symmetric monoidal on the category whose morphisms are coupling-respecting maps.*

The proof reduces to the corresponding statements for the algebra of open dynamical systems [15], [16], [19]: coupling data are exactly the “wires” of a wiring diagram in those formalisms, and our coupling-respecting morphisms are the morphisms of that wiring-diagram-algebra.

Computational implication. $\boxtimes_\kappa$ is the formal foundation for “modular substrates”: the biological–atmospheric–cultural composite is built as $(\mathbf{X}_{\text{bio}} \boxtimes_{\kappa_{12}} \mathbf{X}_{\text{atm}}) \boxtimes_{\kappa_{23}} \mathbf{X}_{\text{LLM}}$ rather than as a flat Cartesian product. Each component substrate keeps its own kernel; coupling kernels are independent code paths.

Computational regimes.

We say a substrate is *lattice-like* if $X = \prod_{p \in L} \Sigma$ for some discrete lattice L and per-site alphabet Σ (cellular automata, Lenia, NCA), *particle-like* if $X = \bigsqcup_n (\mathbb{R}^d)^n$ (Particle Lenia, swarms, Avida-with-positions), *graph-like* if X is a function space on a graph that may itself evolve, or *discrete-program-like* if elements of X are programs in some instruction set (Avida, LLM-policy substrates). $\boxtimes_\kappa$ allows arbitrary heterogeneous compositions, e.g. lattice $\boxtimes_\kappa$ particle.

D. Entities

The framework requires a precise notion of **entity**: many ALife applications have communities of substrate sites (a Lenia blob, a flock, a colony of Avida programs) acting as a single locus of variation and selection. We adopt an

information-theoretic, substrate-agnostic definition that generalises the individuality framework of Krakauer *et al.* [17] and connects cleanly — but not exclusively — to the closure-operator viability machinery of §III-F.

The definition has two reader-facing jobs. First, it identifies the part of the substrate that currently supports the thing we want to call an entity. Second, it states the scale at which we will describe that thing. A flock may be supported by many particles but described by a centroid; a program may be supported by one memory region and described by its raw instruction sequence.

Definition III.4 (Entity). Let $\mathbf{X} = (X, \Phi, \mu_0)$ be a substrate, and let $X = \prod_{p \in P} X_p$ be a (possibly trivial) decomposition of the state space into per-site factors indexed by a set P of *substrate sites*. An *entity* at time t is a pair $e = (S, \pi)$ consisting of:

- 1) a *supporting set* $S \subseteq P$ of substrate sites, and
- 2) a *scale projection* $\pi : \prod_{p \in S} X_p \rightarrow Y_S$ to a measurable feature space Y_S ,

such that, with $\mathbf{x}_S^t := (\mathbf{x}_p^t)_{p \in S}$ and $\mathbf{x}_S^t := (\mathbf{x}_p^t)_{p \notin S}$,

(E1) Persistence (information closure). There exists $\tau \geq 1$ such that

$$I(\pi(\mathbf{x}_S^{t+\tau}); \pi(\mathbf{x}_S^t) | \mathbf{x}_S^t) \geq I(\pi(\mathbf{x}_S^{t+\tau}); \mathbf{x}_S^t) - \varepsilon,$$

for a specified tolerance $\varepsilon \geq 0$. Thus, most information predictive of π 's future is internal to S , conditional on the outside.

(E2) Boundedness. $1 \leq |S| < \infty$ (or S has finite positive measure under a fixed reference measure on P).

(E3) Compositionality. Hierarchical entities are admitted: if $e_1 = (S_1, \pi_1), \dots, e_k = (S_k, \pi_k)$ are entities with disjoint S_i , then $e^* = (\bigcup S_i, (\pi_1, \dots, \pi_k))$ is an entity provided (E1) holds at the joint scale.

The pair (S, π) separates *what supports the entity* (the sites S) from *what scale of description we are committing to* (the projection π). Trivial $\pi = \text{id}$ recovers raw-state entities (e.g. an Avida program). Non-trivial π recovers macroscopic entities such as *Lenia Orbium* (where π might be a centroid-and-mass map) or particle-system flocks (where π might be the empirical centroid and inertia tensor).

(E1) closely resembles the “informational closure” criterion of Krakauer *et al.* [17] and Bertschinger *et al.* [20]. It is *not* identical to the closure-operator characterisation of viable supporting sets in §III-F — that machinery operates on the structure of the supporting set S alone, while (E1) operates jointly on (S, π) . The relationship is left as a conjecture (§V-A.1).

The scale projection π explicitly anticipates the coarse-graining structure of causal emergence (§III-I): every entity carries with it a candidate macroscale at which one may compute effective information.

Computational implication. Detecting entities reduces to (i) a search over candidate supporting sets S (e.g. connected components above a threshold for *Lenia*, or autopoietic preimages for Game-of-Life gliders), and (ii) a search over scale projections

π . Both steps are embarrassingly parallel across candidates; see §III-L.

E. Variation

A **variation operator** is a measurable map

$$V : X \times \Theta \rightarrow \mathcal{P}(X),$$

where Θ is a parameter space (mutation rate, recombination probabilities, gradient step size, prompt-perturbation distribution). Stochastic, deterministic, and gradient-based variation are unified by allowing V to be a Markov kernel; deterministic variation is the Dirac case. In ordinary simulator language, V is the mutation, recombination, rewrite, or optimisation step that proposes candidate next entities.

Instances span Gaussian/uniform mutation, bit-flip, k -point and uniform recombination, evolution strategies [21], gradient descent (treated as deterministic V), prompt mutation for LLM-policies, foundation-model-guided variation [12], and Intrinsically Motivated Goal Exploration Processes [22].

A **variation channel** is a pair (V_c, Θ_c) identified with one inheritance lineage (genetic, cultural, etc.). Coupled inheritance channels require an operation richer than a tensor product. We define a **channel composition operator** $\diamond$:

Definition III.5 (Channel composition). Given channels V_1 and V_2 that act on substrate components $\mathbf{X}_1$ and $\mathbf{X}_2$ of a coupled substrate $\mathbf{X}_1 \boxtimes_\kappa \mathbf{X}_2$, and a *coupling rule* ρ specifying how sampling in one channel may condition sampling in the other (e.g. cultural variation may depend on genetic state [23]), the composed channel $V_1 \diamond_\rho V_2$ is the kernel

$$\begin{aligned} (V_1 \diamond_\rho V_2)((x_1, x_2), (\theta_1, \theta_2)) \\ = \int V_1(x_1, \theta'_1) V_2(x_2, \theta'_2) \rho(d\theta'_1, d\theta'_2 | x_1, x_2, \theta_1, \theta_2). \end{aligned}$$

Trivial ρ (independent product) recovers tensor-product variation $V_1 \otimes V_2$ as a special case; non-trivial ρ supports gene–culture co-evolution [23], niche construction [24], and stigmergic coupling [25].

Computational implication. $\diamond_\rho$ is implemented as a single fused kernel that emits both channels’ offspring jointly when they are coupled, and as two independent kernels when they are not — the same modularity argument as for $\boxtimes_\kappa$.

Topology-mediated population variation.

The lifted operator $\tilde{V}_T$ in §III-B requires a *population-level* variation kernel $V_T : \mathcal{M}_+(X) \times X \rightarrow \mathcal{M}_+(X)$, derived from the per-entity variation operator V of §III-E together with the interaction topology T_t (§III-G). Define V_T as follows. Let $\mathcal{E}_t$ be the entity set induced by μ at substrate state x (the entities $e = (S, \pi)$ admitted by Definition III.4). Let T_t be a hypergraph on $\mathcal{E}_t$ with hyperedge set $E(T_t)$ and hyperedge weights w_e . For each hyperedge $e \in E(T_t)$ involving entities $(e_1, \dots, e_r)$, choose a parameter $\theta_e = g(e_1, \dots, e_r)$ via a measurable rule g (default: θ_e depends only on the focal entity, recovering independent mutation; non-trivial g encodes recombination, prompt-conditioning, etc.). Then

$$V_T(\mu; x) = \sum_{e \in E(T_t)} w_e V(\text{state}(e), \theta_e),$$

interpreted as a finite positive measure on X (the channel composition operator $\diamond_\rho$ of §?? extends this to multi-channel variation under the same construction). When $E(T_t)$ is the discrete set of singletons $\{\{e_i\}\}$ with unit weights, V_T collapses to the per-entity push-forward of V along μ . The expression $\tilde{V}_T(x, \mu) = (x, V_T(\mu; x))$ in §III-B now type-checks.

F. Viability

A **viability filter** is a measurable map $F : \mathcal{M}_+(X) \rightarrow \mathcal{M}_+(X)$ that selects a sub-population of the post-variation population. It is the rule that says which proposed entities remain in the run and with what weight. We characterise the principal proposals in the literature in terms of closure-operator structure on supporting sets.

Four formalisms.

- **Markov-blanket viability** [26], [27]. An entity’s supporting set is required to admit a sparse-coupling decomposition into internal/blanket/external states. Caveats apply [28], [29]; see §III-J.2.
- **Autopoietic closure** [30]–[33]. The supporting set must be closed under the network of constitutive reactions/transitions; e.g. a Game-of-Life glider’s supporting set is closed under the GoL preimage relation restricted to the entity’s reaction network.
- **RAF set viability** [34], [35]. A reaction set R' is a Reflexively Autocatalytic Food-generated (RAF) set if every reaction in R' is catalysed by something in R' (or in the food set) and every reactant is producible from the food set within R' .
- **Minimal Criterion Coevolution** [36]; see also [4]. An entity passes the filter iff it satisfies a minimal qualifying predicate (e.g. “solves at least one maze”), with co-evolutionary feedback supplying the predicate’s content.

Closure-operator interpretation.

The four viability formalisms suggest, but do not yet establish, a common closure-operator interpretation. We therefore treat the unification as Conjecture C1 (§V-A.1), with partial supporting evidence presented there. **Until C1 is settled, the four formalisms are not in general substitutable in code:** a simulator that swaps Markov-blanket viability for autopoietic closure must verify that the substitution preserves the predicates the rest of the system relies on. The implementation checklist of §III-L.8 therefore requires the viability formalism to be named explicitly, not merely as “ $F : \mathcal{M}_+(X) \rightarrow \mathcal{M}_+(X)$ ”.

Hierarchical viability.

Following (E3) of §III-D, we propose a **hierarchical viability filter** F^* that accepts a composite entity e^* iff each component entity e_i satisfies its filter *and* the joint scale projection $(\pi_1, \dots, \pi_k)$ also satisfies a top-level filter. This is the obvious move; verifying that the four formalisms behave correctly under hierarchical composition is open (§V-A.2).

G. Interaction Topology

The **interaction topology** specifies *who interacts with whom*: it is a graph (or, more generally, a hypergraph or

simplicial complex) on entities, possibly time-varying. It is not the material geometry itself. A substrate may be a square lattice, while the interaction topology may be the graph of gliders that collide, agents that exchange messages, or particles within a k -nearest-neighbour radius.

Interaction topology vs substrate topology.

Two distinct notions of topology must be kept separate:

- **Substrate topology.** The intrinsic geometry of the substrate state space X (e.g. the lattice L of a CA; the manifold structure of $\mathbb{R}^d$ for particle systems). This is a property of $\mathbf{X}$ in §III-C and *does not appear in the substrate tuple* — it is whatever geometry the dynamics Φ already presupposes.
- **Interaction topology.** A causal coupling structure T on entities: T is a (possibly weighted, possibly directed, possibly time-varying) graph $T_t \subseteq \mathcal{E}_t \times \mathcal{E}_t$, where $\mathcal{E}_t$ is the entity set at time t .

The two coincide only in homogeneous lattice CA where every entity is one site. Otherwise they differ — a *Lenia Orbium* and a meteorological field couple via a specific spatial overlap relation that is neither the lattice topology of the CA nor the trivial complete graph on entities.

Interaction topology definition.

An **interaction topology** is a measurable family $T = (T_t)_{t \in \mathbb{N}}$ where each T_t is a hypergraph on $\mathcal{E}_t$ together with edge weights in some semiring R (typically $\mathbb{R}_{\geq 0}$ or $\{0, 1\}$). Instances include lattice neighbourhoods, range-based interactions on $\mathbb{R}^d$ (k NN, ε -ball), social graphs (sampled or fixed), pairwise game graphs (donor–recipient pairings as in [37]), and stigmergic indirect-coupling topologies in which interactions pass through the substrate rather than directly between entities [25].

Multi-regime topologies.

When several coupling regimes coexist (a colony with both within-colony chemical signalling and between-colony pheromone trails), T is no longer a single graph but a **layered multiplex**: $T = \bigsqcup_\ell T^{(\ell)}$ with non-negative layer weights (α_ℓ) summing to one (so the total interaction load per tick is preserved). Variation V_T on a multiplex topology is the corresponding *mixture* of layer kernels, $V_T(\mu; x) = \sum_\ell \alpha_\ell V_{T^{(\ell)}}(\mu; x)$, when channels are independent, or the $\diamond_\rho$ -composition (cf. §??) when they are not. The mixture form guarantees that $V_T(\mu; x) \in \mathcal{M}_+(X)$ whenever each layer kernel is.

H. Observer–Evaluator

The observer–evaluator O acts on trajectories. In the stateless case, we model it as a functional

$$O : Z^w \rightarrow \mathbb{R}^k,$$

where $Z = X \times \mathcal{M}_+(X)$ is the joint state space of §III-B and $w \in \mathbb{N} \cup \{\infty\}$ is a fixed *window length* (number of consecutive ticks O consumes). Components of the output are interpreted as *observed quantities* (novelty, complexity, learnability, value). Practical observers use finite w ; $w = \infty$ recovers the long-trajectory limit. The window length w is

part of the observer specification and is reported in the implementation checklist (§III-L.8).

Many observers used in novelty search and Quality-Diversity algorithms are stateful rather than purely functional. To accommodate these cases without adding a sixth framework slot, we allow O to carry an observer state $a_t \in A_O$:

$$O_s : Z^w \times A_O \rightarrow \mathbb{R}^k \times A_O.$$

The state a_t may contain an archive of previous behaviours, a descriptor map, a learned embedding model, a reference distribution, or a saturation criterion. The public output of the observer remains the score vector in $\mathbb{R}^k$; the internal state records how novelty, diversity, or boundedness is computed across time.

Thus the observer is not another hidden force in the default simulator loop. It is the measurement apparatus: it watches the history and reports novelty, complexity, value, effective information, or another chosen quantity.

Three families.

- **Density-distance observers.** Score a window of trajectory by a divergence $D(\mu_{[t-w,t]} \parallel \mu_{\text{ref}})$ — e.g. KL divergence to a uniform reference, or Wasserstein distance to an archive.
- **FM-embedding observers.** Embed states or trajectories with a foundation model (vision-language or otherwise) and score in embedding space, as in ASAL [13] and OMNI-EPIC [14].
- **Information-theoretic observers.** Mutual information, transfer entropy, integrated information, or — at the macroscale — effective information (used as the basis of §III-I emergence).

Observer-relative novelty as divergence.

Following novelty search [4], learnability-oriented open-endedness [18], and the Ω metric [38], we treat observer-relative open-endedness as an assessment made by a specified observer. A strong version says that a system exhibits **observer-relative open-endedness** w.r.t. O iff the observer's score is unbounded along the trajectory (in a suitably regularised sense — typically learnable novelty, as in [18]). A weaker version reports sustained novelty or increasing functional information over a finite horizon but does not infer unboundedness. The distinction is important: a bounded system may still show emergence, adaptation, or increasing functional information, while failing a strict open-endedness test.

This is **distinct** from the mathematical definition of emergence we develop in §III-I.

Adjacent possible as a special case.

Tria *et al.* [39] and the Pólya-urn-with-triggering family yield Heaps' law for novelty rate $\sim t^\beta$ with $\beta < 1$. This is recovered when O maintains an archive of distinct π -equivalence classes seen so far (with an open-ended class set) and emits the count of new classes per tick. First-occurrence triggers in the variation operator are realised by an *archive feedback* $\rho_O : O\text{-state} \rightarrow \Theta$ that re-parameterises V on first-occurrence events — formally, the same ρ -coupling

construction of §?? with O 's archive state playing the role of an additional channel. This keeps V 's signature intact while permitting history-dependent triggering.

I. Emergence

This paper uses a single mathematical definition of emergence: **Hoel-style causal emergence**, anchored on **effective information** [2], [3], [40], [41]. This definition is not derived from the closure-operator machinery of §III-F or the observer-relative divergence of §III-H; instead, the relationship between those components and causal emergence is posed as an open problem.

Effective information.

Let $M = (X, P)$ be a discrete-time, finite-state Markovian system on state space X with transition matrix P , $P_{ij} = \Pr[x^{t+1} = j \mid x^t = i]$. Let $\mathbf{H}_{\max}$ denote the uniform distribution over X (the **intervention distribution**: each cause is intervened upon equiprobably, à la Pearl).

Definition III.6 (Effective information). The *effective information* of M is

$$\text{EI}(M) := I_{\mathbf{H}_{\max}}(X^t; X^{t+1}) = \frac{1}{|X|} \sum_{i \in X} D_{\text{KL}}(P_{i,\cdot} \parallel \bar{P}),$$

where $\bar{P} = |X|^{-1} \sum_i P_{i,\cdot}$ is the marginal effect distribution under uniform interventions. Equivalently, $\text{EI}(M) = \mathbf{H}(\bar{P}) - |X|^{-1} \sum_i \mathbf{H}(P_{i,\cdot})$, the difference between effect-distribution entropy and average row entropy. EI admits a decomposition into determinism (low row-entropy) and non-degeneracy (broad effect distribution); see [2].

Causal emergence.

Let $\Pi : X \twoheadrightarrow X_\Pi$ be a **coarse-graining** (a measurable partition of X into macrostates, with X_Π denoting the quotient set), inducing a macroscale Markov chain $M^\Pi = (X_\Pi, P^\Pi)$ where P^Π is the lifted transition matrix obtained by averaging micro-transitions weighted by a macro-state distribution (see [40] for the canonical choice).

Definition III.7 (Causal emergence). The *causal emergence* of M at coarse-graining Π is

$$\text{CE}(M, \Pi) := \text{EI}(M^\Pi) - \text{EI}(M).$$

The system M is *causally emergent* iff

$$\sup_{\Pi} \text{CE}(M, \Pi) > 0,$$

i.e. there exists a coarse-graining at which the effective information strictly exceeds that of the microscale.

Emergence of substrate-agnostic systems.

For a system $\mathcal{S}$ in the sense of §III-B, write $M(\mathcal{S})$ for the Markov chain on the joint state space $Z = X \times \mathcal{M}_+(X)$ induced by the composite update rule $\tilde{S}_F \circ \tilde{V}_T \circ \tilde{\Phi}$. (For substrate-as-population paradigms in which μ is implicit in x , $M(\mathcal{S})$ reduces to a chain on X .) We say $\mathcal{S}$ *exhibits causal emergence at scale Π* iff $\text{CE}(M(\mathcal{S}), \Pi) > 0$, and that $\mathcal{S}$ is *causally emergent* iff this holds for some Π , i.e. if $\sup_{\Pi} \text{CE}(M(\mathcal{S}), \Pi) > 0$ (Definition III.7).

This definition is by construction:

- **Intrinsic.** EI is computed from the transition matrix; no external observer enters its definition. (Contrast §III-H.)
- **Coarse-graining-based.** The macroscale is a partition Π of Z ; not a closure condition on supporting sets. (Contrast §III-F.)
- **Non-trivial.** Whether $\sup_{\Pi} \text{CE} > 0$ holds for given S is a non-trivial property requiring construction of Π . (See §V-A.4 for the open problem.)

Extensions and known limitations.

Φ ID-based emergence [42], [43] replaces partition-based coarse-grainings with arbitrary supervenient features $V_t = f(X_t)$, defined causally emergent iff V has unique predictive information about X^{t+1} that no microscopic component has. This is feature-agnostic but methodologically more demanding.

Comolatti and Hoel [41] show that causal emergence is widespread across causation measures (over a dozen popular formalisms): emergence is not an artefact of using EI specifically.

Limitations. The intervention distribution $\mathbf{H}_{\max}$ is conventional, not unique; results are sensitive to the choice [44]. Sup-over- Π is only computable for small $|X|$; tractable algorithmic relaxations exist [40], [45] but do not exhaust the supremum.

Computational implication. EI is a per-Markov-chain reduction; it composes with the population-level μ_t but does not parallelise across the population — it is computed once per chain, after Φ, V, F have produced P . Coarse-graining search is itself parallel: candidate partitions can be evaluated independently.

J. Evolution, Culture, and Active-Inference Models

Two channels, transmission, niche construction.

A standard cultural-and-genetic system is

$$S = (\mathbf{X}_g \boxtimes_{\kappa} \mathbf{X}_c, V_g \diamond_{\rho} V_c, F, T, O)$$

where $\mathbf{X}_g$ carries the genetic state, $\mathbf{X}_c$ the cultural state, κ encodes how genetic state shapes cultural transmission and vice versa (gene–culture co-evolution [23]), and the niche-construction map of [24] is implemented as a coupling kernel from population state to environmental sub-substrate. The Donor Game with LLM agents [37] is the discrete instance: $\mathbf{X}_c$ is the prompt-conditioned policy state, V_c is prompt mutation, T_t is the random donor–recipient pairing, F is fitness-proportional reseeding.

The free energy principle as a descriptive lens.

We treat the FEP [26], [27] as a **descriptive lens** rather than a derivation: when a system S admits a Markov-blanket decomposition that is preserved under Φ , then internal-state dynamics can be *parameterised* as variational free-energy minimisation. We do not claim this decomposition is universal; technical critiques [28], [29] show that sparse coupling does not imply Markov-blanket structure in general, and Friston’s reply [46] sharpens but does not resolve the challenge. The FEP is a useful *parameterisation* of viability and observer

slots when applicable; it is not a general theorem of substrate-agnostic systems. Morphogenesis-as-Bayesian-inference [47] is the canonical biological application.

K. Representative Model Instantiations

- **Lenia** [48]. $X = [0, 1]^L$ on a 2D toroidal lattice L , Φ a kernel-convolution-and-growth update.
- **Flow-Lenia** [49], [50]. Adds local mass conservation and parameter localisation, enabling multi-species cohabitation.
- **Particle Lenia** [51]. Energy-based Lagrangian particle reformulation.
- **Neural Cellular Automata** [52]. Φ is a small neural network applied per-cell.
- **Differentiable Logic CA** [53], building on differentiable logic gate networks [54], [55]. Discrete-state CA with differentiable training via logic-gate softmaxes.
- **Particle / swarm.** Boids, Particle Life. Particle-like substrate; interaction topology a k -NN or ε -ball.
- **Avida** [56]; original system [57]. Discrete-program substrate.
- **LLM-policy substrate.** Prompt-conditioned policies as elements of X ; V is prompt mutation or LLM-mediated re-writing [12], [37].

Component-Level Instances.

- **Variation:** Gaussian/uniform mutation, bit-flip, k -point recombination, ES [21], gradient descent (deterministic V), prompt mutation, IMGEP [22], and FM-guided exploration [13], [14].
- **Viability:** Markov-blanket viability; autopoietic closure [31]–[33]; RAF detection [34], [35]; MCC [36]; novelty-archive viability [4]; Quality-Diversity / MAP-Elites filters; POET viability [58], [59].
- **Interaction topology:** lattice neighbourhoods; range-based on $\mathbb{R}^d$; static social graphs; dynamic pairing graphs (Donor Game [37]); stigmergic indirect coupling.
- **Observer:** density-distance archives (novelty search); CLIP-embedding observers (ASAL [13]); Hughes–Dennis novelty-and-learnability [18]; Ω metric [38].

Default choices for canonical substrate families.

The framework is permissive but a simulator author needs starting points. Table II covers the four substrate families introduced in §III-C; deviations should be reported in the implementation checklist (§III-L.8).

These are starting points only; an author who departs from them must specify the alternative explicitly in the checklist of §III-L.8.

Worked Composite Example.

We instantiate every typed object of the scaffold for a Flow-Lenia coupled to an LLM-naming agent. The point is not the system’s empirical interest but to exhibit a fully-specified instance: a simulator author wishing to implement the framework can follow this template line-for-line and obtain a typed iteration rule.

TABLE II
DEFAULT SLOT CHOICES FOR CANONICAL SUBSTRATE FAMILIES.

Slot	Lattice-like	Particle-like	Graph-like	Program-like
π (entity scale)	connected component above threshold $\rightarrow$ centroid + mass	flock identifier $\rightarrow$ centroid + inertia tensor	community identifier $\rightarrow$ membership vector	$\pi = \text{id}$ (raw program)
V default	parameter-localised Gaussian on kernel weights	Gaussian on per-particle parameters	edge rewiring + vertex resampling	bit-flip + instruction insertion
F default	autopoietic closure on the local reaction network	spatial-cohesion threshold (cluster persists $\geq \tau$ ticks)	RAF detection on edge-catalysis graph	MCC predicate (executes a non-trivial program)
T default	Moore neighbourhood on L	k -NN on positions	the underlying graph itself	trivial / topology supplied by interaction protocol
O default	Ω metric on attractors	density-distance archive on $\mathbb{R}^d$	graph-level invariants + embedding novelty	fitness + FM-embedding novelty
Window	$w \approx \tau_{\text{relax}}$	$w \approx \tau_{\text{relax}}$	$w \approx \text{diam}(G)$	$w \approx K$ instructions

a) *Substrate*: $\mathbf{X} = \mathbf{X}_{\text{FL}} \boxtimes_{\kappa} \mathbf{X}_{\text{LLM}}$, where

- $\mathbf{X}_{\text{FL}} = ([0, 1]^L, \Phi_{\text{FL}}, \mu_0^{\text{FL}})$ with $L = (\mathbb{Z}/H\mathbb{Z}) \times (\mathbb{Z}/W\mathbb{Z})$ a toroidal lattice and Φ_{FL} the Flow-Lenia kernel-convolution-and-growth update of [49];
- $\mathbf{X}_{\text{LLM}} = (\Sigma^*, \text{id}, \mu_0^{\text{LLM}})$ with Σ^* the space of token sequences and the LLM substrate dynamics taken to be the identity (the LLM is queried, not autonomously evolved);
- coupling $\kappa = (\kappa_{\text{FL} \rightarrow \text{LLM}}, \kappa_{\text{LLM} \rightarrow \text{FL}})$ with $\kappa_{\text{FL} \rightarrow \text{LLM}}(x_{\text{FL}}) = \text{render}(x_{\text{FL}})$ (a text-prompt context drawn from the rendered frame) and $\kappa_{\text{LLM} \rightarrow \text{FL}}(x_{\text{LLM}})$ a per-creature scalar viability bonus indexed by the entity's support S each tick.

b) *Entities*: For each connected component $S \subseteq L$ of $\{p \in L : x_p^{\text{FL}} \geq \theta\}$ above mass-threshold θ , define

$$\pi_{\text{FL}}((x_p)_{p \in S}) = \left(\frac{1}{m} \sum_{p \in S} p x_p, m \right),$$

$$m := \sum_{p \in S} x_p,$$

giving an $\mathbb{R}^2 \times \mathbb{R}_{\geq 0}$ -valued centroid-and-mass scale projection. LLM agents are atomic entities: S a singleton, $\pi = \text{id}$.

c) *Variation*: $V = V_{\text{FL}} \diamond_{\rho} V_{\text{LLM}}$ with

- $V_{\text{FL}}(x; \theta) = \mathcal{N}(x, \sigma^2 I)$ applied to the parameter-localised Flow-Lenia kernel weights [50];
- $V_{\text{LLM}}(x; \theta)$ resamples each token of x with probability p_{rewrite} from the LLM marginal $q(\cdot | \text{context})$;
- $\rho(d\theta'_{\text{LLM}} | x_{\text{FL}}, x_{\text{LLM}}, \theta_{\text{LLM}})$ re-weights the token marginals proportionally to the recent naming-success score of the named creatures, biasing prompt mutations toward referents that paid off in the last W ticks.

d) *Viability*: $F = F^*$ a hierarchical filter (§III-F):

- each Flow-Lenia entity $e_{\text{FL}} = (S, \pi_{\text{FL}})$ passes iff its support S is closed under the local mass-conservation reaction network — autopoietic closure [31];
- each LLM agent passes iff it has successfully named at least one Flow-Lenia entity in the last W ticks (MCC predicate [36]);
- F^* accepts the joint configuration iff both component filters pass and $|\mathcal{E}_{t+1}^{\text{FL}}| \geq 1$ (a non-extinction top-level predicate).

e) *Interaction topology*: $T_t = T_t^{(1)} \sqcup T_t^{(2)} \sqcup T_t^{(3)}$ a three-layer multiplex with weights $(\alpha_1, \alpha_2, \alpha_3)$: $T_t^{(1)}$ is spatial-overlap on L for Flow-Lenia entities (hyperedges = pairs whose supports intersect within radius r); $T_t^{(2)}$ is uniformly

random pairing of LLM judges and namers; $T_t^{(3)}$ is the bipartite stigmergic layer connecting an LLM agent to each Flow-Lenia entity it has named.

f) *Observer*: $O = (O_{\Omega}, O_{\text{CLIP}})$ with window $w = W$:

- $O_{\Omega}((x_{t-w:t}, \mu_{t-w:t}))$ is the residence-time Ω metric of [38] on the time series of distinct Flow-Lenia attractors;
- $O_{\text{CLIP}}((x_{t-w:t}, \mu_{t-w:t}))$ is the mean cosine novelty of CLIP embeddings of (frame, name) pairs against an archive of size A .

g) *Iteration rule*: $(x_{t+1}, \mu_{t+1}) = (\tilde{S}_{F^*} \circ \tilde{V}_{T_t} \circ \tilde{\Phi})(x_t, \mu_t)$ exactly as in §III-B, with each lifted operator constructed as specified above.

h) *Emergence assessment (open)*: Whether $\sup_{\Pi} \text{CE}(M(S), \Pi) > 0$ holds for this system is a concrete instance of OP1 (§V-A.4).

The scaffold's non-prescriptiveness is now visible: every slot can be ablated independently — replace V_{LLM} with random prompts, swap F^* for Markov-blanket viability, collapse the multiplex T to a complete graph, narrow O to O_{CLIP} alone — and the resulting system remains an instance of the same scaffold. Conformance (§III-L.8) requires only that each ablation reports its choices in the same template.

L. Implementation and Reproducibility

Implementation is part of the methodology because the framework is intended to be reproducible as a simulator scaffold, not only interpreted as a taxonomy. This subsection therefore records the algorithmic control flow, synchronization boundaries, complexity assumptions, and reporting checklist needed to instantiate S in software. No source code is required; the pseudocode records control flow and synchronization points. The five-slot decomposition is preserved: each slot becomes its own kernel.

Inputs and conventions.

Let N be the population size of entities; $|X|$ be the substrate state size (e.g. $H \times W$ for a 2D lattice, or D for an $\mathbb{R}^D$ particle system); c_{Φ}, c_V, c_F be the per-call costs of one substrate update, one variation draw, and one viability evaluation respectively; c_{nn} the cost of a neighbourhood lookup or k -NN query; and P the number of parallel workers (SMs $\times$ warp-equivalents in CUDA terms; XLA threads in JAX terms).

Per-tick algorithm.

At each tick $t \rightarrow t + 1$:

- 1) **Φ -kernel (substrate dynamics).** For each substrate site or particle, apply Φ in parallel.
 - *Lattice / NCA / Lenia / Flow-Lenia*: a local stencil. Each thread computes the update for one site, reading from a halo. Embarrassingly parallel modulo halo exchange.
 - *Particle / swarm*: parallel over particles; each thread queries its neighbourhood via a spatial-hash or BVH index. k -NN is the interaction step.
 - *Discrete-program substrates (Avida-like)*: parallel over programs; each thread executes one program for K instructions.
- 2) **T -kernel (interaction topology lookup).** Build T_t if dynamic. Local stencils need no global structure; k -NN/ ε -ball lookups require a spatial index, which itself parallelises in $O(\log |X|)$ depth. Social graphs are gather operations indexed by an adjacency list.
- 3) **V -kernel (variation).** For each entity i in parallel, draw mutated offspring. Per-entity variation is embarrassingly parallel; recombination requires an additional gather step indexed by mating pairs (selected via T).
- 4) **F -kernel (viability).** For each entity in parallel, evaluate viability predicate. Fitness-proportional selection requires a parallel prefix sum (scan) over the population — a $\log N$ -depth synchronisation barrier.
- 5) **μ -reduction.** Update population measure; for mass-conserving substrates (Flow-Lenia) include a global mass reduction over the lattice — another $O(\log |X|)$ -depth scan.
- 6) **O -kernel (observer).** Optionally: archive update, divergence computation, or FM embedding. k -NN novelty is all-to-all (or approximate-NN) across the archive; FM embedding is an embarrassingly parallel batch forward pass.

The scaffold's products $S_F \circ V_T \circ \Phi$ map onto **kernels with explicit synchronisation only at boundaries between slots**. Within a slot, parallelism is per-site (for Φ, F) or per-entity (for V , novelty).

Single-threaded complexity.

Per tick, ignoring constants:

$$T_1(N, |X|) = O(|X|c_\Phi + |E_T|c_{nn} + Nc_V + Nc_F + c_O(N, |X|)),$$

where $|E_T|$ is the number of edges in T_t (e.g. $O(|X|)$ for lattices with bounded-degree neighbourhoods, $O(Nk)$ for k -NN, $O(N^2)$ for complete pairwise interaction) and c_O depends on observer family.

Many-threaded complexity.

With P parallel workers, ideal-case (no contention) per-tick

cost is

$$T_P(N, |X|) = O\left(\frac{|X|c_\Phi + |E_T|c_{nn}}{P} + \frac{Nc_V + Nc_F + c_O(N, |X|)}{P} + (\log N + \log |X|)c_{\text{sync}}\right),$$

where the log terms come from population-level reductions (μ -mass conservation; fitness-proportional selection prefix-sum) and from spatial-index construction. In the lattice case with $|E_T| = \Theta(|X|)$ and stencil cost $c_\Phi = \Theta(1)$, this collapses to $O(|X|/P + N/P + \log |X|)$.

Communication patterns.

- **Embarrassingly parallel.** Per-site Φ on a lattice (with halo); per-entity V when channels are independent; per-entity F when the predicate is local; FM-embedding observers (batch forward).
- **Local stencils.** Lenia/NCA/Flow-Lenia Φ -kernels; halo width = kernel radius. Standard CUDA shared-memory stencil pattern.
- **Population-level reductions.** Mass conservation in Flow-Lenia ($O(\log |X|)$); fitness-proportional selection prefix sum ($O(\log N)$); Ω -metric residence-time reductions.
- **All-to-all.** k -NN observers against archive (all current entities vs all archive entries); naive $O(NM)$, approximate-NN $O(N \log M)$.
- **Gather operations.** Social-graph indexed lookups; T -mediated mating pair gather. Bandwidth-limited rather than compute-limited.

Random-number considerations.

Each parallel worker requires an independent stream. Counter-based RNGs such as Philox 4×32 -10 [60] are appropriate: stream (i, t) is a key-counter pair ((global seed, i), t), allowing reproducible per-worker streams without locks or pre-computation. JAX's `jax.random.split` API instantiates exactly this pattern.

Modularity and code-reuse benefits.

Each of the five slots becomes a separate kernel. Implementations are:

- **Reusable across substrates:** the V -kernel for Gaussian mutation is identical whether the substrate is Lenia or a Boids swarm; only the input shape differs.
- **Reusable across viability formalisms:** Markov-blanket, autopoietic, RAF, and MCC viability all type as $\mathcal{M}_+(X) \rightarrow \mathcal{M}_+(X)$; switching between them is a single function pointer, not a rewrite.
- **Reusable across observers:** density-distance, FM-embedding, and EI-based observers all take a trajectory and return a vector in $\mathbb{R}^k$.

This is the pragmatic payoff of the abstract slot decomposition.

Implementation Checklist.

A reproducible $\mathcal{S}$ -implementation reports, for each slot:

- 1) Substrate $\mathbf{X} = (X, \Phi, \mu_0)$, including $\boxtimes_k$ -decomposition if composite.

- 2) Entity definition (supporting set determination; scale projection π).
- 3) Variation channels and any $\diamond_p$ couplings.
- 4) Viability filter family (named, not merely typed) and predicates; if departing from the default of §III-K.2, justify.
- 5) Interaction topology including any multiplex layers and weights (α_ℓ).
- 6) Observer family, window length w , and state variables, if any (archive size, embedding model, reference distribution, etc.).
- 7) Descriptor space, archive update rule, saturation criterion, and boundedness/open-endedness test used by the observer.
- 8) Iteration order. Default: $\tilde{S}_F \circ \tilde{V}_T \circ \tilde{\Phi}$. Any deviation (e.g. $\tilde{V}_T \circ \tilde{S}_F \circ \tilde{\Phi}$ for asynchronous selection) is stated explicitly.
- 9) RNG provenance per kernel (counter-based scheme, (seed, worker id, t) key triple).
- 10) Per-tick wall-clock and theoretical complexity.
- 11) Pseudocode or executable specification for initialization, tick update, observer-state update, stopping rule, and outputs.

a) Minimal pseudocode: A conformant implementation supplies at least the following control-flow template, specialised to its chosen slots:

- 1) Initialise (x_0, μ_0) , topology state, observer state a_0 , and RNG keys.
- 2) For $t = 0, \dots, T - 1$, compute $x' = \Phi(x_t)$ and push μ_t forward if entities are explicit.
- 3) Build or update T_t and draw the varied population $\mu' = V_{T_t}(\mu_t; x')$.
- 4) Apply the viability filter $\mu_{t+1} = F(\mu')$ and set $x_{t+1} = x'$.
- 5) Evaluate $(o_t, a_{t+1}) = O_s((x, \mu)_{t-w:t+1}, a_t)$, update any archive, and test the stopping or saturation criterion.
- 6) Return trajectory summaries, observer outputs, archive state, and reproducibility metadata.

b) Conformance: Two implementations are *conformant* with respect to the framework when they (i) agree on the typing of each slot (§III-B–§III-H); (ii) report all eleven items of the checklist with values, not just slot names; (iii) preserve the iteration order of item 8 or explicitly declare a variant; and (iv) specify the entity-detection rule consistently with Definition III.4. Conformance does *not* require choice of the same fitness landscape, observer family, or coarse-graining II; those are user-level choices. It does require that a comparison across two implementations be a comparison of choices, not of unstated defaults.

IV. RESULTS

V. DISCUSSION

A. Conjectures and Open Problems

This section consolidates *all* claims in the scaffold that we have *not* proven. They are stated as conjectures or open

problems to distinguish them from definitions and established propositions.

Conjecture (closure-operator unification of viability).

Conjecture (C1: closure-operator unification of viability). *Each of the four viability formalisms — Markov-blanket viability, autopoietic closure, RAF set membership, and MCC — corresponds to a closure operator $c_\bullet$ on a partial order of supporting sets (suitable to the formalism), in the sense that an entity passes the filter iff its supporting set is closed under $c_\bullet$.*

a) Status: Partial evidence:

- *Autopoiesis.* Beer’s formalisation of Game-of-Life gliders [31], [32] explicitly uses preimage-closure on the reaction network — this is a genuine closure operator on the reaction-set lattice.
- *RAF.* The RAF property is defined exactly as “ R' is closed under reflexive autocatalysis and food-generation”; the maxRAF is the largest closed set. Hordijk & Steel’s polynomial-time algorithm is a closure-iteration. This case is essentially proven.
- *Markov blanket.* The statistical decomposition is *not obviously* a closure operator. Sparse-coupling implies blanket structure only under additional conditions [28], [61]. C1 in this case is genuinely open.
- *MCC.* The minimal criterion is a predicate, not obviously a closure. Reformulating it as “smallest set of solver-and-problem pairs satisfying mutual qualification” gives a closure-like structure but the equivalence is unproven.

A proof of C1 would unify the four formalisms abstractly; a refutation (e.g. a counterexample for the Markov-blanket case) would reveal that “viability” is genuinely heterogeneous.

Conjecture (hierarchical viability composition).

Conjecture (C2: hierarchical viability composition). *Under the entity composition rule (E3), each viability formalism in C1 lifts to hierarchical entities such that F^* -validity at the composite level implies F -validity of each component. Equivalently: if each component closure operator agrees on its component supporting set, the composite is closed under the joint closure operator.*

b) Status: Open. Beer’s autopoietic gliders compose correctly under simple unions; RAF sets do not in general (a union of two RAFs is not necessarily an RAF without their joint catalyses), giving a partial counterexample for one direction.

Conjecture (observer-relative novelty as divergence).

Conjecture (C3: observer-relative novelty as divergence). *All three observer families in §III-H.1 — density-distance, FM-embedding, information-theoretic — are instances of a single divergence-functional schema $O[\mu] = D(\mu \parallel \mu_{\text{ref}})$ for a context-dependent reference μ_{ref} .*

c) Status: Plausible by inspection but unproven in generality.

Open problem (causal emergence of scaffolded systems).

Open Problem (OP1: causal emergence of scaffolded systems). *Find substrate–viability pairs $(\mathbf{X}, F)$ for which one can prove $\sup_{\Pi} \text{CE}(M(S), \Pi) > 0$ for the scaffolded system S .*

This manuscript does *not* attempt a proof of OP1. Concrete candidate systems include:

- *Flow-Lenia substrates with Markov-blanket viability filters.* Are these emergent in the sense of Definition III.7?
- *Differentiable Logic CA implementing Conway’s Game of Life.* Israeli and Goldenfeld [62], [63] showed Class III/IV ECAs admit non-trivial coarse-grainings; does CE follow?
- *POET / Enhanced POET coevolved environments and agents.* Is there a coarse-graining at which environment–agent dynamics achieve higher EI than the joint microscale?

OP1 is the scaffold’s first concrete open problem.

Open problem (FEP-as-lens scope).

Open Problem (OP2: FEP-as-lens scope). *Characterise precisely the substrate–dynamics pairs for which the FEP is informative as a parameterisation of viability and observer slots, in light of [28], [29] and Friston’s reply [46].*

Open problem (interaction-topology expressiveness).

Open Problem (OP3: interaction-topology expressiveness). *Is the multiplex-with- $\diamond_p$ formalism in §III-G.3 universal, in the sense that every “natural” multi-coupling-regime ALife system can be expressed as a multiplex with countably many layers? A categorical formulation would express this in terms of the operad of wiring diagrams [15].*

B. Scope and Commitments

The scaffold **commits to** the five-slot decomposition, the substrate composition operator $\boxtimes_\kappa$, the variation-channel composition operator $\diamond_p$, the entity definition (Def. III.4), the strict separation of substrate topology from interaction topology, the canonical iteration order $\tilde{S}_F \circ \tilde{V}_T \circ \tilde{\Phi}$ (variants must be declared), and the Hoel-style definition of emergence (Def. III.7).

The scaffold **stays neutral on** the choice of fitness landscape, the choice of interaction-topology family, the choice of observer family (we offer three but commit to none), the metaphysical status of agency, the specific closure-operator structure of viability formalisms (left as Conjecture C1), and — critically — the relationship between causal emergence and the rest of the framework (left as OP1).

C. Limitations and Submission Positioning

The framework is intended to support comparison rather than to impose a single theory of agency, fitness, or open-endedness. Its main advantage is that a model can vary one slot while holding the others fixed, making ablations and cross-substrate comparisons explicit. Its main limitation is that several unifications remain conjectural: C1–C3 and OP1–OP3 identify where the framework still requires either proof, counterexample, or domain-specific instantiation.

For IEEE submission, the manuscript should therefore be read as a methodology and framework paper rather than an empirical benchmark paper. The representative instances and worked composite example demonstrate expressibility and implementation structure; they do not by themselves establish a new empirical state of the art. A venue that expects experimental validation may require a follow-on section with simulator results, ablations, or a quantitative comparison across at least two substrate families.

VI. CONCLUSION

REFERENCES

- [1] C. Gershenson, “Emergence in artificial life,” *Artificial Life*, vol. 29, no. 2, pp. 153–167, 2023.
- [2] E. P. Hoel, L. Albantakis, and G. Tononi, “Quantifying causal emergence shows that macro can beat micro,” *Proceedings of the National Academy of Sciences*, vol. 110, no. 49, pp. 19 790–19 795, 2013.
- [3] E. P. Hoel, “When the map is better than the territory,” *Entropy*, vol. 19, no. 5, p. 188, 2017.
- [4] J. Lehman and K. O. Stanley, “Abandoning objectives: Evolution through the search for novelty alone,” *Evolutionary Computation*, vol. 19, no. 2, pp. 189–223, 2011.
- [5] R. P. Wiegand, “Preliminary analysis of simple novelty search,” *Evolutionary Computation*, vol. 32, no. 3, pp. 249–273, 2024.
- [6] M. Faldor and A. Cully, “Toward artificial open-ended evolution within Lenia using quality-diversity,” 2024.
- [7] R. Ishizawa, H. Sato, and K. Takadama, “Novelty-based multi-objectivization for unbounded search space optimization,” *Artificial Life and Robotics*, vol. 30, pp. 383–397, 2025.
- [8] J. M. Borg, A. Buskell, R. Kapitany, S. T. Powers, E. Reindl, and C. Tennie, “Evolved open-endedness in cultural evolution: A new dimension in open-ended evolution research,” *Artificial Life*, vol. 30, no. 3, pp. 417–438, 2024.
- [9] R. M. Hazen and M. L. Wong, “Open-ended versus bounded evolution: Mineral evolution as a case study,” *PNAS Nexus*, vol. 3, no. 7, p. pgae248, 2024.
- [10] S. Stepney, “Towards origins of virtual artificial life: An overview,” *Philosophical Transactions of the Royal Society B*, vol. 380, no. 1936, p. 20240298, 2025.
- [11] E. Hemberg, S. Moskal, and U.-M. O’Reilly, “Evolving code with a large language model,” *Genetic Programming and Evolvable Machines*, vol. 25, p. 21, 2024.
- [12] J. Lehman, J. Gordon, S. Jain, K. Ndousse, C. Yeh, and K. O. Stanley, “Evolution through large models,” in *Handbook of Evolutionary Machine Learning*. Springer, 2024, pp. 331–366.
- [13] A. Kumar, C. Lu, L. Kirsch, Y. Tang, K. O. Stanley, P. Isola, and D. Ha, “Automating the search for artificial life with foundation models (ASAL),” 2024, also *Artificial Life* 31(3), 368, 2025.
- [14] M. Faldor, J. Zhang, A. Cully, and J. Clune, “OMNI-EPIC: Open-endedness via models of human notions of interestingness with environments programmed in code,” 2024, iCLR 2025.
- [15] D. Vagner, D. I. Spivak, and E. Lerman, “Algebras of open dynamical systems on the operad of wiring diagrams,” *Theory and Applications of Categories*, vol. 30, pp. 1793–1822, 2015.
- [16] P. Schultz, D. I. Spivak, and C. Vasilakopoulou, “Dynamical systems and sheaves,” *Applied Categorical Structures*, vol. 28, pp. 1–57, 2020.
- [17] D. Krakauer, N. Bertschinger, E. Olbrich, J. C. Flack, and N. Ay, “The information theory of individuality,” *Theory in Biosciences*, vol. 139, pp. 209–223, 2020.
- [18] E. Hughes, M. D. Dennis, J. Parker-Holder, F. Behbahani, A. Mavalankar, Y. Shi, T. Schaul, and T. Rocktaschel, “Position: Open-endedness is essential for artificial superhuman intelligence,” in *Proceedings of the 41st International Conference on Machine Learning*, ser. PMLR, vol. 235, 2024, pp. 20 597–20 616.
- [19] D. J. Myers, “Double categories of open dynamical systems,” in *Electronic Proceedings in Theoretical Computer Science*, vol. 333, 2021, pp. 154–167.
- [20] N. Bertschinger, E. Olbrich, N. Ay, and J. Jost, “Autonomy: An information-theoretic perspective,” *Biosystems*, vol. 91, no. 2, pp. 331–345, 2008.
- [21] T. Salimans, J. Ho, X. Chen, S. Sidor, and I. Sutskever, “Evolution strategies as a scalable alternative to reinforcement learning,” 2017.

- [22] S. Forestier, R. Portelas, Y. Mollard, and P.-Y. Oudeyer, "Intrinsically motivated goal exploration processes with automatic curriculum learning," *Journal of Machine Learning Research*, vol. 23, no. 152, pp. 1–41, 2022.
- [23] R. Boyd and P. J. Richerson, *Culture and the Evolutionary Process*. University of Chicago Press, 1985.
- [24] F. J. Odling-Smee, K. N. Laland, and M. W. Feldman, *Niche Construction: The Neglected Process in Evolution*, ser. Monographs in Population Biology. Princeton University Press, 2003, vol. 37.
- [25] P.-P. Grasse, "La reconstruction du nid et les coordinations interindividuelles chez *bellicositermes natalensis* et *cubitermes* sp. la theorie de la stigmergie," *Insectes Sociaux*, vol. 6, pp. 41–80, 1959.
- [26] K. Friston, "Life as we know it," *Journal of the Royal Society Interface*, vol. 10, no. 86, p. 20130475, 2013.
- [27] M. Kirchhoff, T. Parr, E. Palacios, K. Friston, and J. Kiverstein, "The markov blankets of life: Autonomy, active inference and the free energy principle," *Journal of the Royal Society Interface*, vol. 15, no. 138, p. 20170792, 2018.
- [28] M. Aguilera, B. Millidge, A. Tschantz, and C. L. Buckley, "How particular is the physics of the free energy principle?" *Physics of Life Reviews*, vol. 40, pp. 24–50, 2022.
- [29] M. Biehl, F. A. Pollock, and R. Kanai, "A technical critique of some parts of the free energy principle," *Entropy*, vol. 23, no. 3, p. 293, 2021.
- [30] H. R. Maturana and F. J. Varela, *Autopoiesis and Cognition: The Realization of the Living*. Reidel, 1980.
- [31] R. D. Beer, "The cognitive domain of a glider in the game of life," *Artificial Life*, vol. 20, no. 2, pp. 183–206, 2014.
- [32] —, "Characterizing autopoiesis in the game of life," *Artificial Life*, vol. 21, no. 1, pp. 1–19, 2015.
- [33] —, "An investigation into the origin of autopoiesis," *Artificial Life*, vol. 26, no. 1, pp. 5–22, 2020.
- [34] W. Hordijk and M. Steel, "Detecting autocatalytic, self-sustaining sets in chemical reaction systems," *Journal of Theoretical Biology*, vol. 227, no. 4, pp. 451–461, 2004.
- [35] W. Hordijk, J. I. Smith, and M. Steel, "Algorithms for detecting and analysing autocatalytic sets," *Algorithms for Molecular Biology*, vol. 10, p. 15, 2015.
- [36] J. C. Brant and K. O. Stanley, "Minimal criterion coevolution: A new approach to open-ended search," in *Proceedings of GECCO 2017*, 2017, pp. 67–74.
- [37] A. Vallinder and E. Hughes, "Cultural evolution of cooperation among llm agents," 2024.
- [38] A. J. Lopez-Diaz, P. J. Rivera Torres, G. L. Febres, and C. Gershenson, "Characterizing open-ended evolution through undecidability mechanisms in random boolean networks," 2025, preprint dated late 2025.
- [39] F. Tria, V. Loreto, V. D. P. Servedio, and S. H. Strogatz, "The dynamics of correlated novelties," *Scientific Reports*, vol. 4, p. 5890, 2014.
- [40] B. Klein and E. Hoel, "The emergence of informative higher scales in complex networks," *Complexity*, vol. 2020, p. 8932526, 2020.
- [41] R. Comolatti and E. Hoel, "Causal emergence is widespread across measures of causation," 2022.
- [42] F. E. Rosas, P. A. M. Mediano, H. J. Jensen, A. K. Seth, A. B. Barrett, R. L. Carhart-Harris, and D. Bor, "Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data," *PLoS Computational Biology*, vol. 16, no. 12, p. e1008289, 2020.
- [43] P. A. M. Mediano, F. E. Rosas, A. I. Luppi, H. J. Jensen, A. K. Seth, A. B. Barrett, R. L. Carhart-Harris, and D. Bor, "Greater than the parts: A review of the information decomposition approach to causal emergence," *Philosophical Transactions of the Royal Society A*, vol. 380, no. 2227, p. 20210246, 2022.
- [44] J. Dewhurst, "Causal emergence from effective information: Neither causal nor emergent?" *Thought: A Journal of Philosophy*, vol. 10, no. 3, pp. 158–168, 2021.
- [45] R. Griesenow, B. Klein, and E. Hoel, "Finding the right scale of a network: Efficient identification of causal emergence through spectral clustering," 2019.
- [46] K. Friston, "Very particular: Reply to "how particular is the physics of the free energy principle?"", *Physics of Life Reviews*, vol. 41, pp. 58–60, 2022.
- [47] F. Kuchling, K. Friston, G. Georgiev, and M. Levin, "Morphogenesis as bayesian inference: A variational approach to pattern formation and control in complex biological systems," *Physics of Life Reviews*, vol. 33, pp. 88–108, 2020.
- [48] B. W.-C. Chan, "Lenia: Biology of artificial life," *Complex Systems*, vol. 28, no. 3, 2019.
- [49] E. Plantec, G. Hamon, M. Etcheverry, P.-Y. Oudeyer, C. Moulin-Frier, and B. W.-C. Chan, "Flow-lenia: Towards open-ended evolution in cellular automata through mass conservation and parameter localization," in *Proceedings of the 2023 Artificial Life Conference*, 2023, best Paper Award.
- [50] E. Plantec, G. Hamon, M. Etcheverry, B. W.-C. Chan, P.-Y. Oudeyer, and C. Moulin-Frier, "Flow-lenia: Emergent evolutionary dynamics in mass conservative continuous cellular automata," *Artificial Life*, vol. 31, no. 2, pp. 228–248, 2025.
- [51] A. Mordvintsev, E. Niklasson, and E. Randazzo, "Particle lenia and the energy-based formulation," 2022, google Research, Self-Organising Systems. [Online]. Available: <https://google-research.github.io/self-organising-systems/particle-lenia/>
- [52] A. Mordvintsev, E. Randazzo, E. Niklasson, and M. Levin, "Growing neural cellular automata," *Distill*, vol. 5, no. 2, p. e23, 2020.
- [53] P. Miotti, E. Niklasson, E. Randazzo, and A. Mordvintsev, "Differentiable logic cellular automata: From game of life to pattern generation," 2025, also Paradigms of Intelligence Team report, March 2025.
- [54] F. Petersen, C. Borgelt, H. Kuehne, and O. Deussen, "Deep differentiable logic gate networks," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.
- [55] F. Petersen, H. Kuehne, C. Borgelt, J. Welzel, and S. Ermon, "Convolutional differentiable logic gate networks," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.
- [56] R. E. Lenski, C. Ofria, R. T. Pennock, and C. Adami, "The evolutionary origin of complex features," *Nature*, vol. 423, pp. 139–145, 2003.
- [57] C. Adami, C. T. Brown, and C. Ofria, "Evolutionary learning in the 2d artificial life system Avida," in *Artificial Life IV: Proceedings of the Fourth International Workshop on the Synthesis and Simulation of Living Systems*. MIT Press, 1994, pp. 377–381.
- [58] R. Wang, J. Lehman, J. Clune, and K. O. Stanley, "Paired open-ended trailblazer (POET): Endlessly generating increasingly complex and diverse learning environments and their solutions," 2019.
- [59] R. Wang, J. Lehman, A. Rawal, J. Zhi, Y. Li, J. Clune, and K. O. Stanley, "Enhanced POET," in *Proceedings of ICML 2020*, ser. PMLR, vol. 119, 2020, pp. 9940–9951.
- [60] J. K. Salmon, M. A. Moraes, R. O. Dror, and D. E. Shaw, "Parallel random numbers: As easy as 1, 2, 3," in *Proceedings of SC11*, 2011.
- [61] C. Heins and L. Da Costa, "Sparse coupling and markov blankets: A comment on "how particular is the physics of the free energy principle?" by aguilera, millidge, tschantz and buckley," *Physics of Life Reviews*, vol. 42, pp. 33–39, 2022.
- [62] N. Israeli and N. Goldenfeld, "Computational irreducibility and the predictability of complex physical systems," *Physical Review Letters*, vol. 92, no. 7, p. 074105, 2004.
- [63] —, "Coarse-graining of cellular automata, emergence, and the predictability of complex systems," *Physical Review E*, vol. 73, no. 2, p. 026203, 2006.